



Faculty of Science and Technology

Programme Structure & Syllabus

For

First Year

Master of Science

Computer Science

(MSC CS)

P-2024

With Effect from Academic Year 2025– 2026

	Authority	Date
Proposed by	Board of Studies in Computer Science	
Approved by	Academic Council, Vishwakarma University, Pune	

Issued by

Chairman – BoS

Dean of Faculty

Director, IQAC

Title: Programme Structure and Syllabus

Form No: IQAC-101

Vision of the University	
Emerge as a Premier University Recognized Internationally for Excellence in Education, Research and Innovation	
Mission of the University	
VU-M1	To impart contemporary transformative education through research and innovation
VU-M2	To develop competent leaders-professionals for life and livelihood
VU-M3	To co-create human and socio-economic capital par excellence
VU-M4	To inculcate life skills and holistic culture appreciating morals and ethics
Values of the University	
Excellence	Transparency
Innovation	Sustainability
Diversity	Responsibility
Adaptability	Compassion

Vision of the Department of Computer Science	
Emerge as renowned research and innovation-oriented department in the field of Information Technology	
Mission of the Department of Computer Science	
M1	To impart industry centric education through research and innovation
M2	To develop skilled professionals to meet global socio-economic challenges
M3	To introduce best practices in software development adhering to moral and ethics
M4	To foster collaborative approach and leadership skills.
Values of the Department of Computer Science	
Integrity	Brightness
Innovation	Diversity
Ethics	Competitive
Leadership	Lucidity

Mapping of Mission Statement of Department to University Mission Statement

Mission Statement	VU-M1	VU-M2	VU-M3	VU-M4
M1	2			
M2		3		
M3			3	
M4				2

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Educational Objectives (PEOs)

PEO No.	Statement
PEO1	Graduates will have a thorough foundational knowledge of the principles and practices of Computer Science, and be proficient in applying them. (Based on Knowledge)
PEO2	To be able to adapt new tools/techniques to effectively contribute as an individual and lead the software development team to provide economically and technically viable solutions to a problem. (Based on Skills)
PEO3	Commit to ethical and moral practices for providing solutions to societal problems through life-long learning (Based on Attitude)

Mapping of Mission Statement of Department to PEOs

Mission Statement	PEO1	PEO2	PEO3
M1	3	2	3
M2	1	3	2
M3	1	2	3
M4	1	3	1

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

1. Programme Outcomes (POs)

PO No.	Statement
PO1	Possess profound knowledge of theory and principles of basic mathematics, statistical techniques and algorithms in the field of computer science.
PO2	Ability to think critically and select economically and technically viable solutions to a given problem.
PO3	Ability to design, develop and deploy solutions for identified problems from diverse domains.
PO4	Provide proof of concept and present novel solutions to challenges encountered by industry.
PO5	Adopt new programming languages, tools, and processes followed in industry
PO6	To work independently or as an effective team member of a software development project.
PO7	Effectively communicate with diverse clients and stakeholders for software project management
PO8	Interpersonal skills to interact with people from diverse cultural backgrounds
PO9	Adhere to the professional, ethical and legal practices followed in the IT industry.
PO10	Possess software project management skills like: cost estimation, effort estimation, scheduling, and team building.

Mapping of PEOs to POs

PEO Number	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
PEO1	3	2	3	2	1	-	-	-	-	-
PEO2	1	3	-	2	3	2	2	1	1	2
PEO3	-	-	2	-	-	1	1	2	3	1

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Specific Outcomes (PSOs)

PSO No.	Statement
PSO1	Choose appropriate data structure, algorithm, and software development paradigm to address real life problems.
PSO2	Apply modern tools and techniques used in Data Science and Game Development.
PSO3	Provide the solutions to computing challenges from multidisciplinary fields.
PSO4	Possess a research and development attitude focusing on knowledge creation and innovation.

Mapping of PEOs to PSOs

PEO Number	PSO1	PSO2	PSO3
PEO1	3	1	2
PEO2	2	3	2
PEO3	1	2	3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Structure

Faculty	Science and Technology	Pattern	2024
Department	Computer Science	Date (w.e.f.)	e.g. 05/08/2024
Programme	M.Sc. – Computer Science		

Semester -I									
Course Type	Course Code	Course Name	Teaching Scheme (Hours/Week)			Credit	Examination Scheme and Marks		
			L	T	P		CIE	ESE	Total
DSC	MSCCS24101	Design and Analysis of Algorithm	3	1	0	4	60	40	100
DSC	MSCCS24102	Advanced Operating System	3	0	0	3	60	40	100
DSC	MSCCS24103	Advanced Operating System Lab	0	0	2	1	15	10	25
DSC	MSCCS24104	Principles of Programming Language	2	0	0	2	50	0	50
DSC	MSCCS24105	Advanced Python Programming	1	0	2	2	50	0	50
DSE	MSCCS24106WD	Advanced Web Technology	3	0	0	3	60	40	100
	MSCCS24106DS	Data Modelling and Visualization							
DSE	MSCCS24107WD	Advanced Web Technology Lab	0	0	2	1	15	10	25
	MSCCS24107DS	Data Modelling and Visualization Lab							
DSC	MSCCS24108	Research Methodology	4	0	0	4	60	40	100
		Total	16	01	06	20	370	180	550
Instructions, if any: 1 Theory or 1 Tutorial Hour = 1 Credit, 2 Practical hours = 1 Credit									

Specialization Elective Tracks

Elective Track	Course Code	Name of Subject
Specialization Elective Track-I (Web Application Development)	MSCCS24106WD	Advanced Web Technology
	MSCCS24107WD	Advanced Web Technology Lab
Specialization Elective Track-II (Data Science)	MSCCS24106DS	Data Modelling and Visualization
	MSCCS24107DS	Data Modelling and Visualization Lab

Semester -II									
Course Type	Course Code	Course Name	Teaching Scheme (Hours/Week)			Credit	Examination Scheme and Marks		
			L	T	P	C	CIE	ESE	Total
DSC	MSCCS24201	Cloud Computing	3	0	0	3	60	40	100
DSC	MSCCS24202	Cloud Computing Lab	0	0	2	1	15	10	25
DSC	MSCCS24203	Software Project Management	3	1	0	4	60	40	100
DSC	MSCCS24204	DevOps Fundamentals	1	1	0	2	50	0	50
DSC	MSCCS24205	NoSQL Databases	2	0	0	2	50	0	50
DSE	MSCCS24206WD	Hypertext Preprocessor	3	0	0	3	60	40	100
	MSCCS24206DS	Artificial Intelligence							
DSE	MSCCS24207WD	Hypertext Preprocessor Lab	0	0	2	1	15	10	25
	MSCCS24207DS	Artificial Intelligence Lab							
OJT	MSCCS24208	Internship	0	0	8	4	60	40	100
Total			12	2	12	20	370	180	550
Instructions, if any: 1 Theory or 1 Tutorial Hour = 1 Credit, 2 Practical hours = 1 Credit									

Specialization Elective Tracks

Elective Track	Course Code	Name of Subject
Specialization Elective Track-I (Web Application Development)	MSCCS24206WD	Hypertext Preprocessor
	MSCCS24207WD	Hypertext Preprocessor Lab
Specialization Elective Track-II (Data Science)	MSCCS24206DS	Artificial Intelligence
	MSCCS24207DS	Artificial Intelligence Lab

Name of Head of Department

Sign:

Date:

Name of Dean

Sign:

Date:

MSCCS24101: Design and Analysis of Algorithm

Course Type	DSC	Semester	I
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hr./Week	Lecture	3	CIE Marks	60
Tutorial	1 Hr./Week	Tutorial	1	ESE Marks	40
Practical/Studio	... Hr./Week	Practical/Studio		Total Marks	100
Total	... Hr./Week	Total			

Course Description

The overall objective of this course is to cater the need of computational field. The content of this course are according to the current trends of research in computer science and requirements of industry expectations. Some courses of this program are exclusively designed towards development of analytical, presentation and personality development skills among the students, through which the students get prepared and trend for building their career in computer science and it related to applied technology, research and development.

Course Outcomes

CO No.	Statement
1	Analyze worst-case running times of algorithms using asymptotic analysis.
2	Compare between different data structures. Pick an appropriate data structure for a design situation.
3	Ability to design algorithms using standard paradigms like: Greedy, Divide and Conquer, Dynamic Programming and Backtracking.
4	Able to Explain the major graph algorithms and Employ graphs to model engineering problems, when appropriate.
5	Able to Compare between different data structures and pick an appropriate data structure for a design situation.

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PSO 1	PSO 2	PSO 3	PSO 4	BTL
CO1	3	1	2								3	2	1	1	1,2
CO2		2	3	1					2	1	2	1	2	3	2,4
CO3				3	2	1		1		3	2	1	4	4	1,3,4
CO4			2	3	1				3		1	3	3	2	5,6
..															

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No I	Introduction to Algorithms	10 Hours	CO	BTL
Algorithm definition and characteristics, Space complexity, Time complexity- worst case, best case, average case, Complexity, asymptotic notation, Recursive and non-recursive algorithms, Sorting algorithms : insertion sort, heap sort, bubble sort, Sorting in linear time: counting sort, concept of bucket and radix sort Searching algorithms: Linear, Binary			1	1,3
Unit No II	Divide and Conquer strategy and Greedy Methods:	14 Hours	2	1,2,4
General method, control abstraction, Binary search, Merge sort, Quick sort, Comparison between Traditional Method of Matrix Multiplication vs. Strassen's Matrix Multiplication, Writing simple algorithm using Divide and conquer strategy: power(x,n), find occurrence of a number from array of N integers, to find minimum from an array, mini-max algorithm, largest number multiplication, simple convex algorithm Greedy Method: Knapsack problem, Job sequencing with deadlines, Minimum-cost spanning trees: Kruskal and Prim's algorithm, Optimal storage on tapes, Optimal merge patterns, Huffman coding, Shortest Path :Dijkstra's Algorithm				
Unit No III	Dynamic Programming and Decrease and conquer:	16 Hours	3	3,4,5
Principle of optimality, Matrix chain multiplication, 0/1 Knapsack Problem i)Merge & Purge ii)Functional Method, Bellman Ford Algorithm, All pairs Shortest Path Floyd-Warshall Algorithm, Travelling Salesperson problem, Longest common subsequence, String editing Decrease and conquer: Definition of Graph Representation of Graph, By Constant - DFS and BFS, Topological sorting, Strongly Connected components and spanning trees, Articulation Point and Bridge edge				
Unit No IV	Backtracking, Branch and Bound Techniques	16 Hours	4	3,4,5
General method, Fixed Tuple vs. Variable Tuple Formulation, n- Queen's problem, Graph colouring problem, Hamiltonian cycle, Sum of subsets Branch and Bound Techniques: Introduction : Branch and bound terms like definition of live node, E-node, Dead node, Least cost (LC) search, Least cost Branch and Bound (LCBB) 0/1 knapsack problem using LCBB method (fixed tuple size), Travelling Salesman problem using LCBB method (variable tuple size)				
Unit No V	Problem Classification	4 Hours	5	2,3
The class of P, NP, NP-hard and NP –Complete, Relationship among P class, NP class, NP-hard and NP –Complete, Cook's theorem				

Textbooks

1	<i>Introduction to Algorithms, Thomas H.Cormen, Charles E.Leiserson, Ronald L. Rivest and Clifford Stein, Third Edition, PHI Learning Private Limited, 2012.</i>
2	<i>Data Structures and Algorithms, Alfred V. Aho, John E. Hopcroft and Jeffrey D. Ullman, Pearson Education, Reprint 2006.</i>
3	<i>Algorithms Design and Analysis, Harsh Bhasin, Oxford university press, 2016.</i>
4	<i>Design and Analysis of Algorithms, S. Sridhar, Oxford university press, 2014.</i>

Reference Books / Journal Articles / Weblink

1	www.w3schools.com
2	www.tutorialspoint.com
3	www.javatpoint.com
4	www.geeksforgeeks.com
5	www.programiz.com
6	www.theserverside.com

MSCCS24102: Advanced Operating System

Course Type	Theory	Semester	3
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hrs./Week	Lecture	3	CIE Marks	60
Tutorial	0 Hrs./Week	Tutorial	0	ESE Marks	40
Practical/Studio	0 Hrs./Week	Practical/Studio	0	Total Marks	100
Total	2 Hrs./Week	Total	3		

Course Objectives

This course enables students to understand advanced operating systems like distributed, real time, network, multiprocessor and mobile operating systems

Course Outcomes

CO No.	At the end of this course students will be able to
1	Gain understanding of basic concepts in operating system and knowledge of database operating systems
2	Learn the architecture and design concepts of distributed operating systems.
3	Study different aspects of network operating systems and multiprocessor systems
4	Understand the working of real time operating systems
5	Study mobile operating systems, web operating systems and using them.

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3	PSO 4	
CO1		1	1		1								1		1	1	1,2
CO2	1	1	1		2								1		1	1	2,3
CO3		1	1		2								1		1	1	2,3
CO4		1	1		2								1		3	1	3,4
CO5	1	1											1		4	2	4,5

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL:Bloom's Taxonomy Level

Course Content

Unit No.01	1 Introduction to Operating System Concepts	9 hours	CO	BTL
Definition and functions of Operating System, Goals of Operating System, File Management: file organization techniques, file allocation strategies, Memory Management, paging techniques, Process Management, Device Management, concurrency control, threading and synchronization			1	1,2
Unit No.02	2 Distributed Operating Systems	9 Hours		
Architecture of distributed operating systems :system architecture types, issues in distributed operating systems, inherent limitation of distributed system, Distributed file system, Distributed mutual exclusion: classification of mutual exclusion algorithms, Lamport's token based algorithm, Suzuki-Kasami's Broadcast algorithm, Raymond's Tree based algorithm, Banker's, Distributed deadlock detection algorithm, Distributed shared memory, Distributed scheduling, examples of Distributed OS			2	3,4
Unit No.03	3 Network and Multiprocessor Operating Systems	9 Hours		
Architecture of Network Operating Systems(NOS), Issues in design of NOS, characteristics of NOS, types of NOS, server and client software for NOS, server hardware, network file system, choosing NOS, Multiprocessor OS:-Introduction, structure of multiprocessor operating system, operating system design issues, threads, test and set instruction, swap instruction, implementation of the process wait , processor scheduling, reliability and fault tolerance, examples of NOS			4	2,3,4
Unit No.04	4 Real Time Operating Systems	9 Hours		
Introduction to Real time systems and Real Time Operating Systems, Characteristics of Real Time operating Systems, Classification of Real Time Operating Systems, Services, structure, goal and feature of RTOS, architecture of RTOS, micro kernels and monolithic kernels, tasks in RTOS, Performance measures, estimating program runtimes, task assignment, scheduling in RTOS, rate monotonic scheduling, priority inversion, task management, inter task communication, applications of various RTOS, examples of RTOS			4	3,4,5
Unit No.05	5 Mobile and Web Operating Systems	9 Hours		
Mobile OS-Structure , platforms and capabilities, Android O.S- architecture, features, kernel design in android OS, application framework, runtime, development tools, File systems on mobile phones, Application development using Android, Web OS-What is web OS,5.5 architecture of web OS, application architecture, APIs, examples of web OS, Introduction to cloud OS			5	4,5,6

Textbooks

1	M Singhal and NG Sivaratri, Advanced Concepts in Operating Systems, Tata McGraw Hill Inc., 2001
2	A.S. Tanenbaum, Distributed Operating system, Pearson Education Asia, 2001. 3. Real -Time Systems, Krishna and Shin, McGraw Hill International

Reference Books / Journal Articles / Weblink

Reference Books	
1	SILBERSCHATZ and P. GALVIN, Operating System Concepts, VI edition, Addison Wesley 2004.
2	https://www.javatpoint.com/operating-system

3	https://www.techtarget.com
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MSCCS24103: Advanced Operating System Lab

Course Type	Practical	Semester	1
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Teaching Scheme		Credits		Examination Scheme	
Lecture	0Hrs./Week	Lecture	0	CIE Marks	25
Tutorial	0 Hrs./Week	Tutorial	0	ESE Marks	25
Practical/Studio	02Hrs./Week	Practical/Studio	1	Total Marks	50
Total	2 Hrs./Week	Total	1		

List Of Experiments
1.Implementing Stack application using Object Oriented tools.
2. Implementing Tree application using Object Oriented tools.
3. Implementing Graph application using Object Oriented tools.
4. Design and test object oriented application based on Structural Design Pattern
5. Design and test object oriented application based on Creational Design Pattern.
6. Design Pattern test object oriented application using Behavioral Design Pattern.
7. Implement web application using HTML, Java script.
8. Implement Web application using Servlets to process client data from server side. .
9. Implement Concurrent applications using object oriented techniques
10. Simulate Object Oriented applications for Event Processing

MSCCS24104: Principles of Programming Language

Course Type	DSC	Semester	I
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Teaching Scheme		Credits		Examination Scheme	
Lecture	02 Hr./Week	Lecture	2	CIE Marks	50
Tutorial	0 Hr./Week	Tutorial	0	ESE Marks	0
Practical/Studio	0 Hr./Week	Practical/Studio	0	Total Marks	50
Total	02 Hr./Week	Total	2		

Course Description

This course, "Principles of Programming Languages," provides a foundational understanding of the core concepts and paradigms of programming languages. Students will explore the historical development, syntax, semantics, and pragmatics of various languages. Key topics include formal language theory, syntax analysis, data types, type systems, and control structures. The course also covers different programming paradigms—imperative, declarative, functional, logic, and object-oriented—along with advanced topics like concurrency and domain-specific languages. By the end, students will be able to evaluate and apply these concepts effectively.

Course Outcomes

CO No.	Statement
1	Understand the historical context and evolution of programming languages and categorize languages based on different paradigms.
2	Analyze and apply formal syntax and semantics to programming language constructs.
3	Understand different data types, type systems, and abstraction mechanisms in programming languages.
4	Apply various control structures and subprogram mechanisms in programming language design.
5	Compare and contrast different programming paradigms and apply paradigm-specific features to solve programming problems.

Mapping of COs to POs and PSOs

CO Number	POs										PSOs			BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PSO1	PSO2	PSO3	
CO1	2										2			1, 2
CO2		3											1	2, 6
CO3			1								3			1, 4
CO4	1	2											1	1
CO5			3										2	2

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Title of Unit: Introduction to Programming Languages	Hours 6	CO	BTL
	- History and Evolution: Early languages, development of modern languages. - Language Paradigms: Imperative, functional, logic, and object-oriented programming languages. - Language Design and Implementation: Syntax, semantics, and pragmatics. - Translators: Compilers, interpreters, and hybrid systems.		1	1, 2
Unit No 2	Title of Unit: Syntax and Semantics	Hours 6		
	- Syntax: Lexical analysis, parsing, and syntax trees. - Formal Language Theory: Grammars, regular expressions, and context-free grammars. - Semantics: Operational, denotational, and axiomatic semantics. - Language Constructs: Expressions, statements, and control structures.		2	1, 2
Unit No 3	Title of Unit: Data Types and Abstraction	Hours 6		
	- Primitive Data Types: Integer, float, boolean, character. - Composite Data Types: Arrays, records, unions. - Abstract Data Types: Encapsulation and modularity. - Type Systems: Static vs. dynamic typing, strong vs. weak typing.		3	1, 2
Unit No 4	Title of Unit: Control Structures and Subprograms	Hours 6		
	- Control Structures: Conditional statements, loops, and control flow mechanisms. - Subprograms: Functions, procedures, and coroutines. - Parameter Passing: By value, by reference, and by name. - Recursion: Concepts and implementation.		4	1, 2
Unit No 5	Title of Unit: Programming Paradigms	Hours 6		
	- Imperative Programming: Variable assignment, loops, and procedure calls. - Functional Programming: First-class functions, higher-order functions, and recursion. - Logic Programming: Facts, rules, and queries. - Object-Oriented Programming: Classes, objects, inheritance, and polymorphism.		5	1, 2

Textbooks

1	<i>Principles of Programming Languages, Gilles Dowek, Springer, 2009</i>
2	Principles of Programming Languages: Design, Evaluation, and Implementation, Bruce J. MacLennan, OUP USA, 3 rd , 1999

Reference Books / Journal Articles / Weblink

1	Principles of Programming Languages, Mike Grant, Zachary Palmer, Scott Smith, Johns Hopkins University, 2011
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MSCCS24105: Advanced Python Programming

Course Type	DSC	Semester	I
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Teaching Scheme		Credits		Examination Scheme	
Lecture	01 Hr./Week	Lecture	01	CIE Marks	50
Tutorial	00 Hr./Week	Tutorial	00	ESE Marks	00
Practical/Studio	02 Hr./Week	Practical/Studio	01	Total Marks	50
Total	01 Hr./Week	Total	02		

Course Description

The Python programming course is essential for the degree program as it provides fundamental to advanced skills vital for contemporary software development. Python's flexibility and rich library ecosystem prepare students for diverse tech careers in areas like data science, web development, and automation. Practical projects and assessments ensure students can apply their knowledge effectively, bridging theory with real-world applications.

Course Outcomes

CO No.	Statement
1	Gain proficiency in Python syntax, basic data structures, control structures (loops, conditionals), and functions/modules to write efficient scripts and programs.
2	Learn to design and implement classes, utilize inheritance and polymorphism, and apply object-oriented concepts to solve complex programming problems.
3	Master functional programming techniques (lambda functions, map, filter, reduce), exception handling, decorators, and generators for efficient and robust coding practices.
4	Familiarize with popular libraries like NumPy and Pandas for data manipulation, Matplotlib for visualization, and Flask/Django for web development to create robust applications.
5	Acquire skills in multithreading, multiprocessing, socket programming, and basics of data science and machine learning using Python libraries like scikit-learn and TensorFlow.

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BT L
	PO 1	PO 2	PO 3	PO 4	PO 5	PO6	PO7	PO8	PO9	P10	PSO1	PSO 2	PSO3	PS O4	
CO1	1	1			1						1				2
CO2		1	1		1						1		1		3
CO3	1	1			1						1				4
CO4	1		1		1							1	1		2
CO5	1	1	1	1	1	1					1	1	1	1	4

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Foundations of Python Programming	2 Hours	CO	BTL
	• Introduction to Python: History, features, and applications in computer science. • Python syntax and basic data structures: Variables, data types, control structures (loops, conditionals). • Functions and modules: Defining functions, scope, built-in and user-defined modules.		1	1,2
Unit No 2	Object-Oriented Programming in Python	3 Hours		
	• Principles of Object-Oriented Programming (OOP): Encapsulation, inheritance, polymorphism. • Classes and objects in Python: Defining classes, instance variables, methods, constructors. • Inheritance and polymorphism in Python.		2	3
Unit No 3	Advanced Python Concepts	3 Hours		
	• Functional programming concepts in Python: Lambda functions, map, filter, reduce. • Exception handling: try-except blocks, raising exceptions. • Decorators and generators: Function decorators, generator functions, yield statement.		3	4,5
Unit No 4	Introduction to Python Libraries and Frameworks	3 Hours		
	• Python libraries and frameworks: NumPy, Pandas, Matplotlib, Flask, Django. • Data manipulation and analysis using NumPy and Pandas.		4	3,4,5
Unit No 5	Advanced Topics in Python	4 Hours		
	• Python for data science and machine learning: Overview, libraries (e.g., scikit-learn, TensorFlow). • Python for Natural Language Processing (NLP): Introduction to NLP tasks, libraries like NLTK, spaCy, etc • Python for Web Scraping and Automation: Using libraries like BeautifulSoup and Selenium, automating tasks and scraping data from websites.		5	6

Textbooks

1	Matthes, Eric. Python Crash Course, 3rd Edition: A Hands-On, Project-Based Introduction to Programming. No Starch Press, 2023.
2	Zelle, John. Python Programming: An Introduction to Computer Science. No Starch Press, 2003.

Reference Books / Journal Articles / Weblink

1	Ramalho, Luciano. Fluent Python: Clear, concise, and effective programming. O'Reilly Media, 2015.
2	https://numpy.org/doc/

3	https://pandas.pydata.org/docs/
4	https://matplotlib.org/stable/contents.html
5	https://flask.palletsprojects.com/
6	Géron, Aurélien. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, tools, and techniques to build intelligent systems, 2nd ed., O'Reilly Media, 2019.
7	NLTK: Official documentation. Retrieved from https://www.nltk.org/
8	spaCy: Official documentation. Retrieved from https://spacy.io/

List of Experiments/ Studio Assignments, etc

Sr. No	Name of Experiment/ Studio Assignment	CO	BTL
1	- Writing Python scripts and basic programs. - Implementing algorithms (e.g., sorting, searching) in Python. - Using Python standard libraries for common tasks (e.g., file handling, string manipulation).	1	3
2	- Implementing class hierarchies and inheritance. - Designing and implementing Python classes to model real-world scenarios. - Applying OOP principles to solve programming problems.	2	3
3	- Writing and using lambda functions, map, filter, and reduce functions. - Implementing custom decorators for logging, performance monitoring. - Creating and using generator functions to process large datasets efficiently.	3	4,5
4	- Performing data analysis and visualization with NumPy and Pandas. - Developing web applications using Flask or Django. - Integrating databases with Python applications (SQLAlchemy, SQLite).	4	4,5
5	- Implementing multithreaded and multiprocessing applications in Python. - Developing networked applications using sockets. - Introduction to data science and machine learning tasks with Python libraries.	5	6

MSCCS24106WD: Advanced Web Technology

Course Type	DSE	Semester	I
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Teaching Scheme		Credits		Examination Scheme	
Lecture	03 Hr./Week	Lecture	3	CIE Marks	60
Tutorial	00 Hr./Week	Tutorial	0	ESE Marks	40
Practical/Studio	00 Hr./Week	Practical/Studio	0	Total Marks	100
Total	03 Hr./Week	Total	3		

Course Description

This course provides a comprehensive introduction to web technologies, covering the fundamental principles of HTML, CSS, and JavaScript for creating and styling web pages. Students will learn to develop interactive web applications using ReactJS, and server-side programming with Node.js and MySQL. The course also explores the MEAN stack (MongoDB, Express.js, Angular, Node.js) for full-stack development. Emphasis is placed on both front-end and back-end development, including web design, form validation, and database connectivity. By the end of the course, students will be equipped with the skills to build dynamic and responsive web applications.

Course Outcomes

CO No.	Statement
1	Understand and apply the fundamentals of web technology and HTML, including structuring HTML documents and styling them with CSS.
2	Develop interactive web applications using JavaScript, focusing on DOM manipulation, event handling, and form validation.
3	Build dynamic and responsive front-end applications using ReactJS, managing state and props, and integrating with backend services.
4	Create and manage server-side applications using Node.js, and perform database operations with MySQL to develop full-stack web applications.
5	Develop full-stack web applications using the MEAN stack, integrating MongoDB, Express.js, Angular, and Node.js to manage the entire application lifecycle.

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Remember
CO2	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Understand
CO3	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Apply
CO4	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Analyze
CO5	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Evaluate
															Create

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Unit No 1	Introduction to web technology	9 Hours	CO	BTL
Introduction to web technology, internet and www, Web site planning and design issues, HTML: structure of an html document , HTML elements: headings, paragraphs, line break, colors & fonts, links, frames, lists, tables, images and forms, Difference between HTML and HTML5, HTML document heading detail. CSS: Introduction to Style Sheets, Inserting CSS in an HTML page, CSS selectors,			1	1,2,3
Unit No 2	JavaScript	9 Hours		
Introduction to JavaScript: JavaScript language - declaring variables, the scope of variables, functions, event handlers (onclick, onsubmit etc.), using JS in an HTML (Embedded, External), Data and objects in JavaScript, regular expressions, exception handling, Document Object Model, Form validation. Simple AJAX application.			2	1,2,3
Unit No 3	Front-end technology:ReactJS	9 Hours		
Introduction - Overview, Basics of JSX, Components and Props, Handling Events and forms, lists, router,Hooks,Integrating react with back end services, Introduction to Redux			3	1,2,3
Unit No 4	Back-end Technologies: Nodes and MySQL	9 Hours		
Node.js : Introduction to node.js,Overview and basics,Modules,HTTP module,File system,Node package Manager(NPM),File system,Debugging,Events,Database Connectivity MySQL: Introduction to jQuery, Overview and basics:Data Types,CRUD(Create,Read,Update,Delete) operations,Joins,node.js and MySQL connectivity			4	1,2,3
Unit No 5	MEAN Stack	9 Hours		
MEAN Stack-Introduction,Overview; Stack components;Mongo DB-Introduction,NoSQL database,MongoDB architecture, CRUD operations with MongoDB;Express.js – Introduction,Routing,Middleware,error handling;CRUD operations with Express and MongoDB,Integrating Angular with Express			5	1,2,3

Textbooks

1	<i>Web Development and Design Foundations with HTML5," Terry Felke-Morris, Pearson, 9th Edition, 2019.</i>
2	<i>Full-Stack Web Development with Vue.js and Node," Aneeta Sharma, Sandeep Kumar Patel, and Amit Kumar, Packt Publishing, 1st Edition, 2018</i>

3	Fullstack React: The Complete Guide to ReactJS and Friends," Anthony Accomazzo, Nathaniel Murray, and Ari Lerner, Fullstack.io, 1st Edition, 2017
4	Learning Web App Development," Semmy Purewal, O'Reilly Media, 1st Edition, 2014

Reference Books / Journal Articles / Databases Documents of Professional Bodies (ASME/ASTEM, etc)

1	JavaScript: The Definitive Guide," David Flanagan, O'Reilly Media, Volume 1, 7th Edition, 2020
2	<i>HTML and CSS: Design and Build Websites," Jon Duckett, Wiley, Volume 1, 1st Edition, 2011</i>
3	<i>Learning React: Modern Patterns for Developing React Apps," Alex Banks and Eve Porcello, O'Reilly Media, Volume 1, 2nd Edition, 2020.</i>
4	Node.js Design Patterns," Mario Casciaro and Luciano Mammimo, Packt Publishing, Volume 1, 2nd Edition, 2016

MSCCS24107WD: Advanced Web Technology Lab

Course Type	DSE	Semester	I
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Teaching Scheme		Credits		Examination Scheme	
Practical/Studio	02 Hr./Week	Practical/Studio	1	CIE Marks	15
				ESE Marks	10
				Total Marks	25

Course Outcomes

CO No.	Statement
1	Understand and apply the fundamentals of web technology and HTML, including structuring HTML documents and styling them with CSS.
2	Develop interactive web applications using JavaScript, focusing on DOM manipulation, event handling, and form validation.
3	Build dynamic and responsive front-end applications using ReactJS, managing state and props, and integrating with backend services.
4	Create and manage server-side applications using Node.js, and perform database operations with MySQL to develop full-stack web applications.
5	Develop full-stack web applications using the MEAN stack, integrating MongoDB, Express.js, Angular, and Node.js to manage the entire application lifecycle.

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Remember
CO2	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Understand
CO3	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Apply
CO4	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Analyze
CO5	-	3	2	2	3	2	-	-	-	1	-	-	1	-	Evaluate
															Create

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Sr. No	Name of Experiment/ Studio Assignment	CO	BTL
1	Create a personal homepage using HTML. Include headings, paragraphs, line breaks, colors, and fonts. Use images and links to create a navigation menu.	1	3
2	Design a multi-page website for a local business. Implement lists, tables, and forms (e.g., contact form) using HTML. Ensure the site is well-structured and uses semantic HTML5 elements.	1	3
3	Apply CSS to style your personal homepage. Use different CSS selectors (element, class, id) to style the headings, paragraphs, and links. Include at least one example of inline, internal, and external CSS.	1	3
4	Create a JavaScript program to validate a web form. Ensure that all required fields are filled out and that the email address is in the correct format before submission.	2	3
5	Develop a simple to-do list application using JavaScript. Allow users to add, delete, and mark tasks as completed. Store the tasks in the browser's local storage.	2	3
6	Build a web page that dynamically updates content based on user input. Use event handlers like onclick and onchange to modify the DOM and display the results.	2	3
7	Create a ReactJS application that displays a list of items. Use components and props to render the items, and state to manage the list. Include functionality to add and remove items.	3	3
8	Develop a ReactJS form with validation. Use hooks to manage form state and validate the input before submission.	3	3
9	Implement a simple routing system in a ReactJS application. Use React Router to navigate between different pages or components within the application.	3	3
10	Set up a basic Node.js server. Create a simple web server that responds with "Hello, World!" when accessed through a browser.	4	3
11	Create a Node.js application that connects to a MySQL database. Implement CRUD operations (Create, Read, Update, Delete) for a simple user management system.	4	3
12	Develop a file upload feature using Node.js. Allow users to upload files to the server and store the file metadata in a MySQL database.	4	3
13	Set up a MongoDB database and perform basic CRUD (Create, Read, Update, Delete) operations. Use MongoDB Shell or a GUI tool to create a database, insert, update, delete, and query documents.	5	3
14	Build a RESTful API using Express.js and MongoDB. Implement endpoints for creating, reading, updating, and deleting resources. Test the API using a tool like Postman.	5	3
15	Integrate an Angular front end with an Express.js back end. Develop a simple application where the Angular front end consumes the RESTful API created with Express and MongoDB to display and manage data.	5	3

Textbooks

1	<i>Web Development and Design Foundations with HTML5," Terry Felke-Morris, Pearson, 9th Edition, 2019.</i>
2	<i>Full-Stack Web Development with Vue.js and Node," Aneeta Sharma, Sandeep Kumar Patel, and Amit Kumar, Packt Publishing, 1st Edition, 2018</i>
3	<i>Fullstack React: The Complete Guide to ReactJS and Friends," Anthony Accomazzo, Nathaniel Murray, and Ari Lerner, Fullstack.io, 1st Edition, 2017</i>
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3	<i>Learning React: Modern Patterns for Developing React Apps," Alex Banks and Eve Porcello, O'Reilly Media, Volume 1, 2nd Edition, 2020.</i>
4	<i>Node.js Design Patterns," Mario Casciaro and Luciano Mammino, Packt Publishing, Volume 1, 2nd Edition, 2016</i>

MSCCS24108: Research Methodology

Course Type	DSC	Semester	I
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Teaching Scheme		Credits		Examination Scheme	
Lecture	4 Hr./Week	Lecture	4	CIE Marks	60
Tutorial	0 Hr./Week	Tutorial	0	ESE Marks	40
Practical/Studio	... Hr./Week	Practical/Studio		Total Marks	100
Total	... Hr./Week	Total	4		

Course Description

To learn critical and creative thinking of models and the components of research with the necessary skills and knowledge to conduct high-quality research in the fields, fostering their ability to contribute significantly to academic and professional communities.

Course Outcomes

CO No.	Statement
1	Gain a solid understanding of research methodology, including objectives, types, and scientific research processes.
2	Learn effective data collection methods such as observation, interviews, surveys, and questionnaire design.
3	Understand criteria and types of sampling designs, including probability and non-probability methods.
4	Apply large and small sample tests, including t-tests, F-tests, and Chi-Square tests for statistical analysis.
5	Improve skills in writing and interpreting research reports, theses, and journal publications.

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PSO 1	PSO 2	PSO 3	PSO 4	BTL
CO1	3	1	2								3	2	1	1	1,2
CO2		2	3	1					2	1	2	1	2	3	2,4
CO3				3	2	1		1		3	2	1	4	4	1,3,4
CO4			2	3	1				3		1	3	3	2	5,6
..															

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No I	Introduction to Research Methodology	12 Hours	CO	BTL
Introduction to Research Methodology: Meaning, Objectives, Motivation in Research, Types, Approaches, Importance, Research Methodology, Scientific Research, Process, Criteria for Good Research, Problems Encountered, Defining Research Problem, Developing Research Proposal, Research Design: Meaning, Need, Features, Important, Six P of Research Data Collection Method- Primary Data- Observation Method, Personal Interview, Telephonic Interview, Mail Survey, Questionnaire Design.			1	1,3
Unit No II	Sampling Design	12 Hours	2	1,2,4
Implications, Steps Criteria for Sampling Procedure, Characteristics of Good Sample Design, Different Types of Sample Design, Different Types of Sample Design- (a) Probability Sampling Like Simple Random, System Random, Systematic Random, Stratified, Cluster.(B) Non Probability Sampling Like Quota, Judgmental, Convenience				
Unit No III	Large and small Sample Test	12 Hours	3	3,4,5
Large Sample Test: Definition of Hypothesis, Basic Concepts- Null Hypothesis and Alternative, The Level of Significance, Type I and Type II Errors, Two Tailed and One Tailed, Power of Test, Testing of Mean, Testing of Differences Between Two Means, Testing of Proportion of Difference Between Two Proportion Limitations of Hypothesis Testing. Small Sample Test: Idea of Degree of Freedom, Test Significance Based Upon T and F Statistic-Testing of Mean, Testing of Difference Between Two Means, Testing of Equality of Variances, Chi-Square Test.				
Unit No IV	CHI-Square Test for Large Samples	12 Hours	4	3,4,5
CHI-Square Test for Large Samples: Definition of Chi-Square, Limitations of Chi-Square Test, and Chi-Square Test As a Test of Goodness of Fit and As a Test of Independence, Yates's Correction and Its Application, Analysis of Variance (ANOVA): Concept, One Way ANOVA, and ANOVA in Test in Latin Square Design.				
Unit No V	Report Writing and Interpretation	12 Hours	5	2,3
Pre-Writing Considerations. Meaning and Technique of Interpretation, Different Types of Report Writing, formats of Report Writing, Thesis Writing, formats of Publication in Research Journals.				

Textbooks

1	1) Kothari .C.R, “Research Methodology-Methods and Techniques”, New Age Publications.
2	2) S.P.Gupta, “Introduction to Mathematical Statics”.
3	3) S.B. Kishor, Bhagyashreevaiddya, “Operation Research”, Das Ganu,

Reference Books / Journal Articles / Weblink

1	D.K.Bhattacharyya, “Research Methodology”, 1st Edition (2003), EBP, New Delhi
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2	Sancheti and Kapoor, “Business Statics”, Sultan Chand and Sons, New Delhi.
3	D.K.Bhattacharyya, “Research Methodology”, 1st Edition (2003), EBP, New Delhi